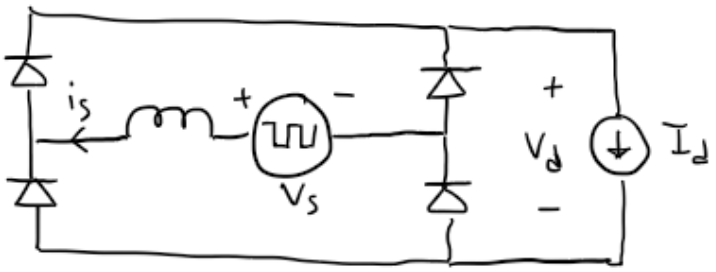
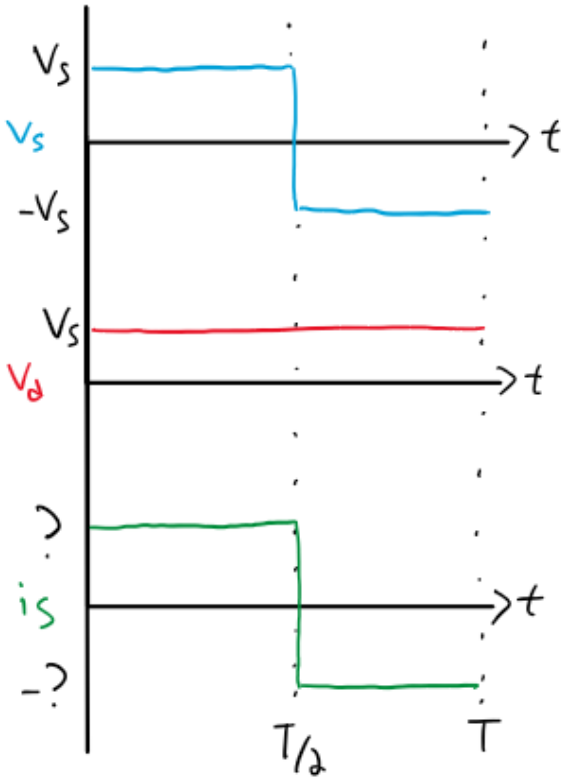
3.16 Square-wave source and leakage inductance



a) $L_c = 0$



What is the magnitude of i_s?

$V_d = V_s$ & $V_R = I_d \cdot R$

$\therefore V_L = 0$: inductor is short-circuit @DC

$\therefore V_R = V_s \quad \therefore I_d = \frac{V_s}{R}$

$|i_s| = \frac{V_s}{R}$

b) $\langle V_d \rangle = V_s$

c) $L_c > 0$

$L_c > 0 \therefore$ commutation interval must exist

↓

Time when all four diodes conduct

During commutation: $V_{L_c} = \pm V_s \therefore \left|\frac{di_s}{dt}\right| = \frac{V_s}{L_c}$

commutation lasts until i_s flips from $+I_d$ to $-I_d$

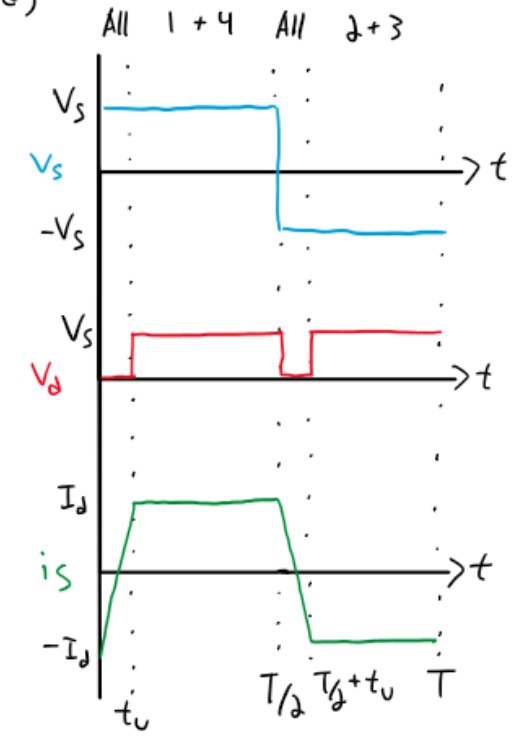
$$2 \cdot I_d = \int_0^{t_u} V_{L_c}\, dt$$

$$2 \cdot I_d = \frac{V_s}{L_c} \cdot \int_0^{t_u} dt$$

$$2 \cdot I_d = \frac{V_s}{L_c} \cdot t_u$$

$$t_u = \frac{2 \cdot I_d \cdot L_c}{V_s}$$

d)



e) Find $\langle V_d \rangle$

$$\langle V_d \rangle = \frac{1}{T/2} \int_0^{T/2} V_d \, dt$$

$$\langle V_d \rangle = \frac{2}{T} \cdot \left[\int_0^{t_u} 0 \cdot dt + \int_{t_u}^{T/2} V_s \, dt \right]$$

$$\langle V_d \rangle = \frac{2}{T} \cdot V_s \cdot \left(\frac{T}{2} - t_u \right)$$

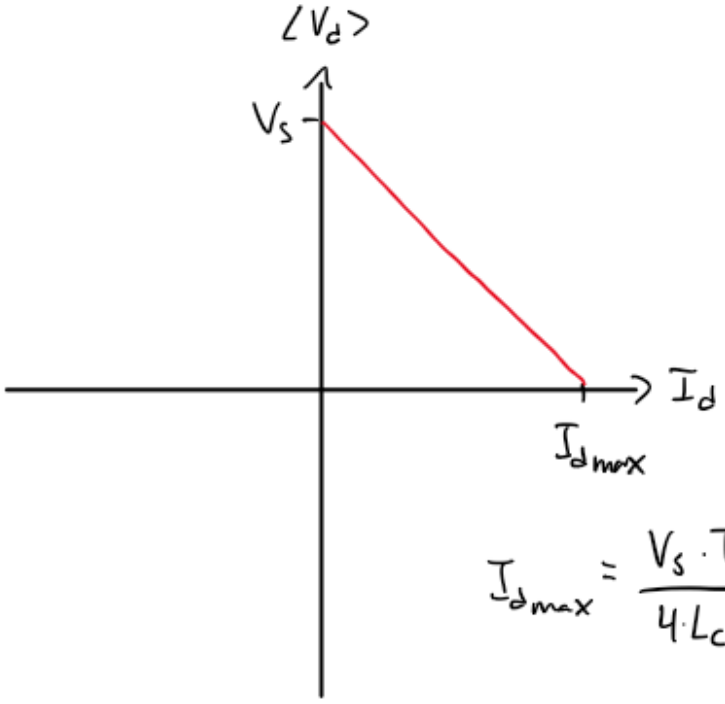
$$\langle V_d \rangle = V_s - \frac{2}{T} \cdot V_s \cdot t_u$$

$$\langle V_d \rangle = V_s - \frac{2}{T} \cdot \frac{2 \cdot I_d \cdot L_c}{V_s} \cdot V_s$$

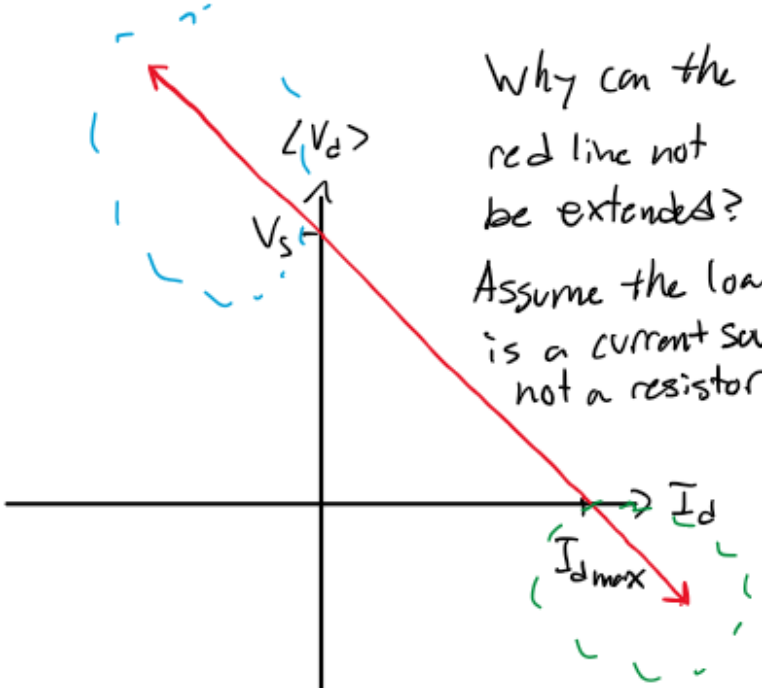
$$\langle V_d \rangle = V_s - \frac{4 \cdot I_d \cdot L_c}{T}$$

f) Sketch the load regulation curve
($\langle V_d \rangle$ verse I_d)

What shape? What are the critical points?



$$I_{d_{max}} = \frac{V_s \cdot T}{4 \cdot L_c}$$



Why can the red line not be extended?

Assume the load is a current source, not a resistor.

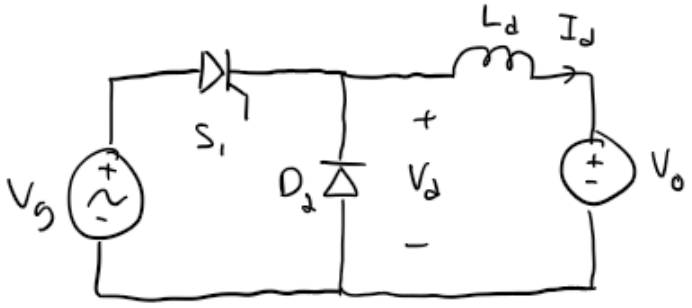
g) t_u & $\langle V_d \rangle$ with

$V_s = 50V$, $T = 10^{-5}s$, $L_c = 10^{-6}H$, $I_d = 20A$

$$t_u = \frac{2 \cdot I_d \cdot L_c}{V_s} = \frac{40 \cdot 10^{-6}}{50} = 8 \cdot 10^{-7}s < \frac{T}{2} \therefore \checkmark$$

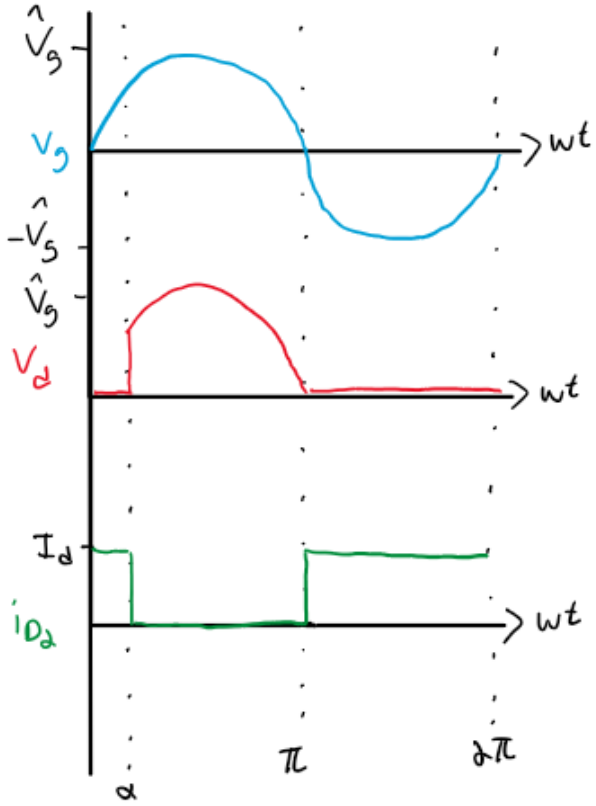
$$\langle V_d \rangle = V_s - \frac{4 I_d L_c}{T} = 50 - \frac{8 \cdot 10^{-5}}{10^{-5}} = 50 - 8 = 42V$$

4.4/4.5 : Half-wave phase control : Diode or thyristor freewheeling



$V_s = \hat{V}_s \sin\omega t$, $i_d(\omega t) = I_d$, S_1 fired @ α $0 \le \alpha \le \pi$

a) Sketch v_d & derive $\langle V_d \rangle$ as function of α +sketch

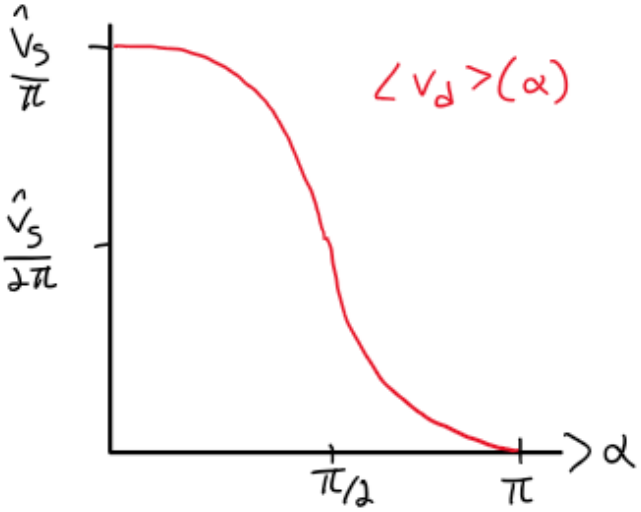


$$\langle V_d \rangle = \frac{1}{2\pi} \cdot \int_0^{2\pi} V_d \, d\omega t$$

$$\langle V_d \rangle = \frac{\hat{V}_s}{2\pi} \cdot \int_\alpha^{\pi} \sin\omega t \, d\omega t$$

$$\langle V_d \rangle = \frac{\hat{V}_s}{2\pi} \cdot (-\cos\pi + \cos\alpha)$$

$$\langle V_d \rangle = \frac{\hat{V}_s}{2\pi} + \frac{\hat{V}_s}{2\pi} \cos\alpha$$



cosine + DC bias

b) Can this circuit be an inverter? Why or why not?

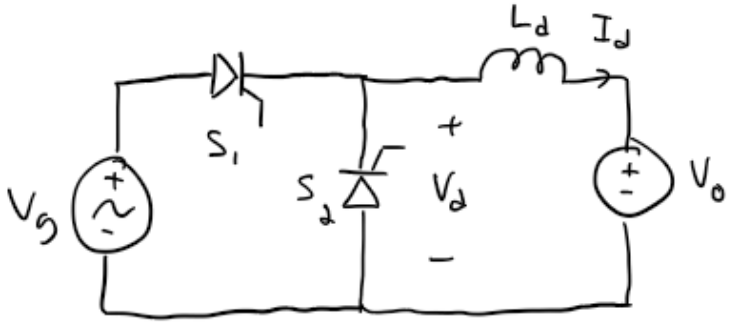
What is needed?

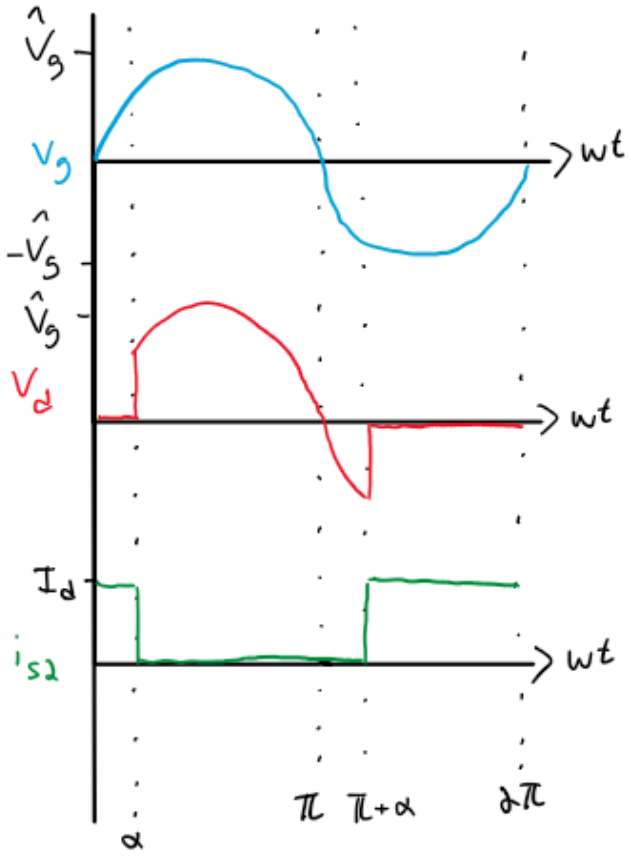
→ V_d to be negative.

Is that possible?

→ No, diode can't block voltage in that direction.

c)


L_d
I_d
S_1
S_2
+
V_d
−
V_g
V_o


$\hat{V}_g$
V_g
wt
$-\hat{V}_s$
$\hat{V}_g$
V_d
wt
I_d
i_{s2}
wt
π
$\pi+\alpha$
2π
α

$$\langle V_d \rangle = \frac{1}{2\pi} \cdot \int_0^{2\pi} V_d \, d\omega t$$

$$\langle V_d \rangle = \frac{\hat{V}_s}{2\pi} \cdot \int_{\alpha}^{\pi+\alpha} \sin\omega t \, d\omega t$$

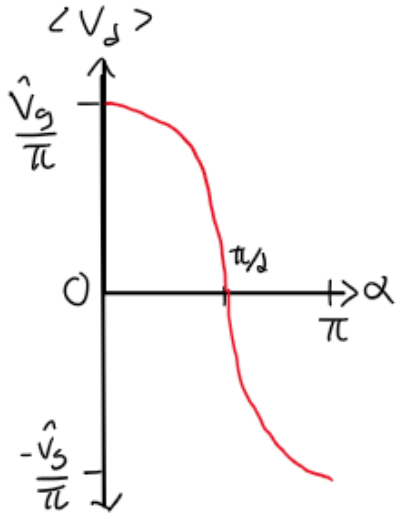
$$\langle V_d \rangle = \frac{\hat{V}_s}{2\pi} \cdot \left(-\cos(\pi+\alpha) + \cos(\alpha) \right)$$

$$\langle V_d \rangle = \frac{\hat{V}_s}{2\pi} \cdot \left(\cos\alpha - \cos(\pi-\alpha) \right)$$

$$\langle V_d \rangle = \frac{\hat{V}_s}{2\pi} \cdot \left(\cos\alpha + \cos(-\alpha) \right)$$

$$\langle V_d \rangle = \frac{\hat{V}_s}{2\pi} \cdot \left(\cos\alpha + \cos\alpha \right)$$

$$\langle V_d \rangle = \frac{\hat{V}_s}{\pi} \cdot \cos\alpha$$



d) For what α range can the circuit behave like an inverter?

$\alpha > \pi/2$: inverter

What are the requirements on the DC voltage source to make this circuit an inverter?

$V_0 < 0$. Why?

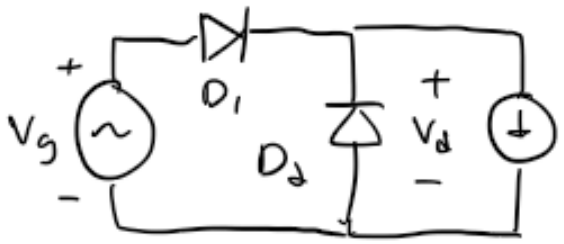
→ I_d must be positive,

so for power balance,

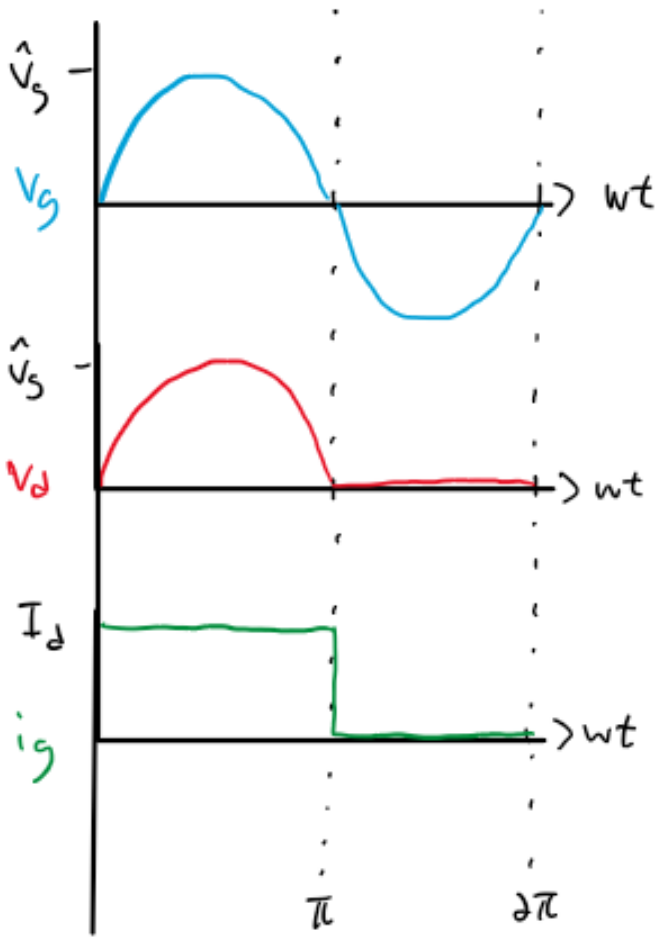
current must go into the lower potential terminal and exit the higher potential terminal.

3.7/3.12: Power factor: half-wave vs. bridge

a) Sketch i_g and find k_p for half-wave.



$$k_p = \cos(\theta_v - \theta_I) \cdot \frac{I_{1rms}}{I_{rms}}$$



In phase, so $\cos(\theta_v - \theta_I) = 1$

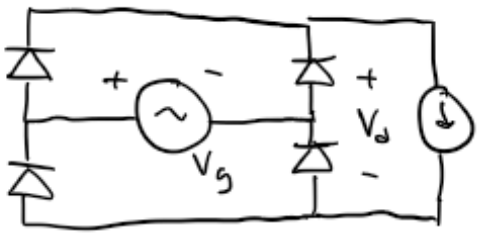
$$k_p = \frac{I_{1rms}}{I_{rms}} \qquad \underset{\text{square wave}}{\hat{I}_1} = \frac{4}{\pi} \cdot \frac{I_d}{2}$$ ← need to get rid of the DC bias of $\frac{I_d}{2}$

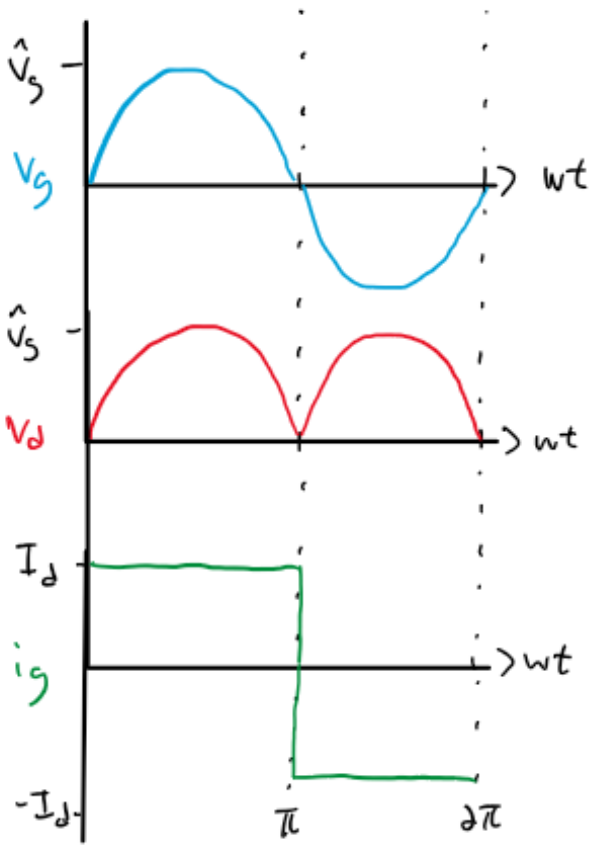
$$I_{1rms} = \frac{2I_d}{\sqrt{2}\,\pi} = \frac{\sqrt{2}\,I_d}{\pi}$$

$$I_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{2\pi} i_d^2\,d\omega t} = \sqrt{\frac{1}{2\pi}\int_0^{\pi} I_d^2\,d\omega t} = \sqrt{\frac{I_d^2 \cdot \pi}{2\pi}} = \frac{I_d}{\sqrt{2}}$$

$$\frac{I_{1rms}}{I_{rms}} = \frac{\frac{\sqrt{2}\,I_d}{\pi}}{\frac{I_d}{\sqrt{2}}} = \frac{2}{\pi} = 0.637$$

b) repeat for full-bridge





$$\hat{I}_{g1} = \frac{4}{\pi} \cdot I_d \qquad I_{1rms} = \frac{2\sqrt{2}}{\pi} \cdot I_d$$

$$I_{1rms} = I_d$$

$$k_p = \frac{\frac{2\sqrt{2} \cdot I_d}{\pi}}{I_d} = \frac{2\sqrt{2}}{\pi} = 0.9$$

c)

half-wave

$$\langle V_d \rangle = \frac{1}{2\pi}\int_0^{2\pi} V_d \, d\omega t$$

$$\langle V_d \rangle = \frac{1}{2\pi}\int_0^{\pi} 120\sqrt{2}\sin\omega t \, d\omega t$$

$$\langle V_d \rangle = \frac{60\sqrt{2}}{\pi}\left(-\cos\pi + \cos 0\right)$$

$$\langle V_d \rangle = \frac{120\sqrt{2}}{\pi}$$

$$I_d = \frac{1000}{\frac{120\sqrt{2}}{\pi}} = 18.51A$$

$$I_{grms} = \frac{I_d}{\sqrt{2}} = 13.1A$$

Apparent power: $120V \cdot 13.1A = 1.57kVA$

full-bridge

$$\langle V_d \rangle = \frac{1}{2\pi}\int_0^{2\pi} V_d \, d\omega t$$

$$\langle V_d \rangle = \frac{1}{\pi}\int_0^{\pi} 120\sqrt{2}\sin\omega t \, d\omega t$$

$$\langle V_d \rangle = \frac{240\sqrt{2}}{\pi}$$

$$I_d = \frac{1000}{\frac{240\sqrt{2}}{\pi}} = 9.255A$$

$$I_{grms} = I_d = 9.26A$$

$120V \cdot 9.26A = 1.11kVA$

d) Why is half-wave power factor worse?

DC current through an AC voltage does not transfer active power.